CS223 Computer Architecture & Organization

Introduction to RISC Instruction Pipeline

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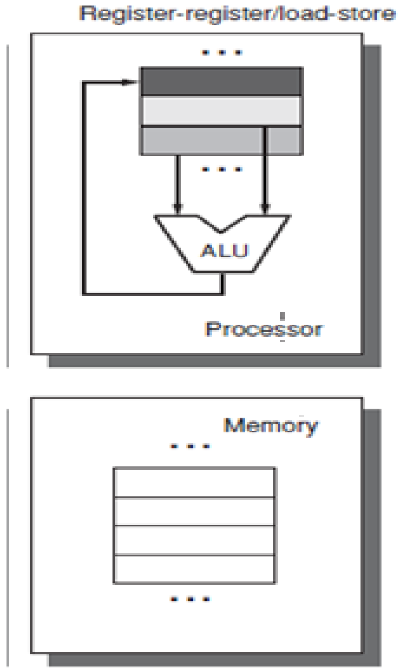
Introduction to MIPS

- ❖Microprocessor without Interlocked Pipelined Stages
- ❖32 registers (32 bit each)
- ❖Uniform length instructions
- ❖RISC- Load store architecture

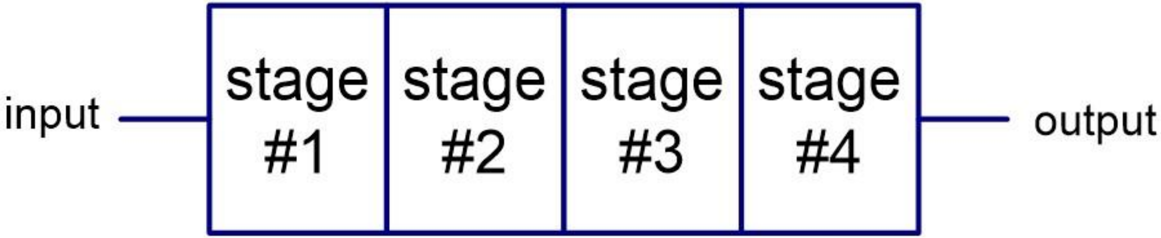


Introduction to MIPS

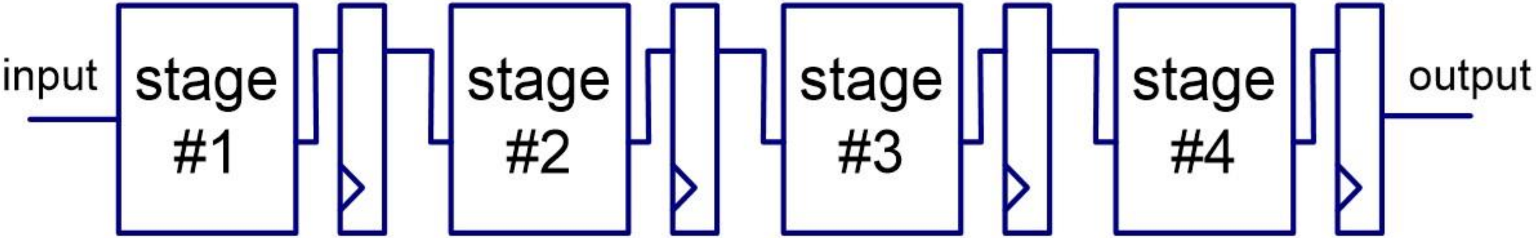
	6 bits	5 bits	5 bits	5 bits	5 bits	6 bits
R:	op	rs	rt	rd	shamt	funct
I:	op	rs	rt	address / immediate		
J:	op	target address				

op: basic operation of the instruction (opcode)
rs: first source operand register
rt: second source operand register
rd: destination operand register
shamt: shift amount
funct: selects the specific variant of the opcode (function code)
address: offset for load/store instructions (+/-2^{15})
immediate: constants for immediate instructions

Combinational Circuit and Pipelining



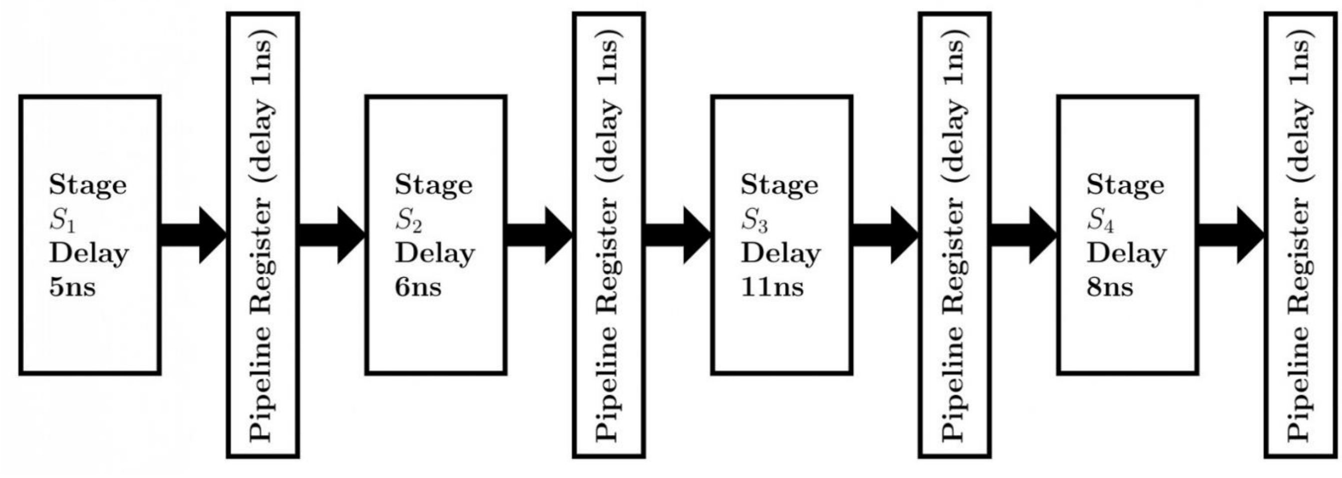
Original combinational circuit



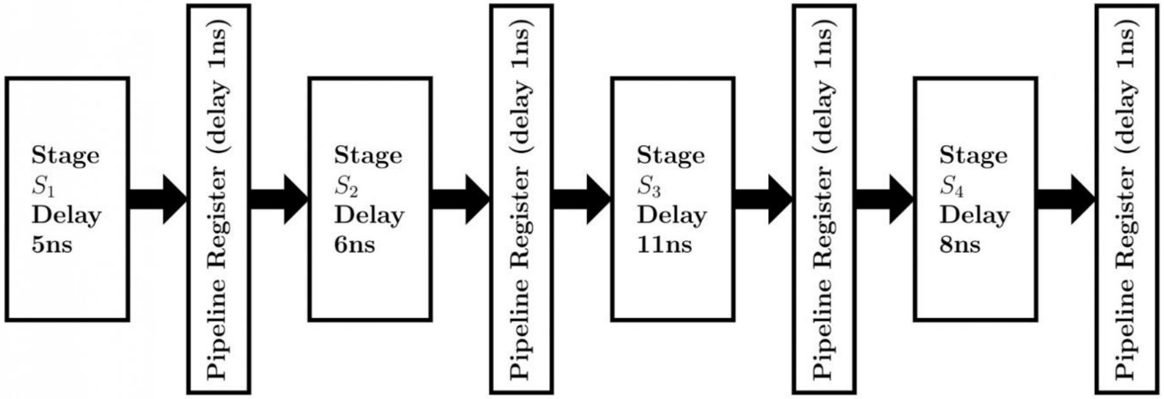
Pipelined circuit

Pipelining Characteristics

- Pipelining partitions the system into multiple independent stages with added buffers between the stages.
- Pipelining can increase the throughout of a system.

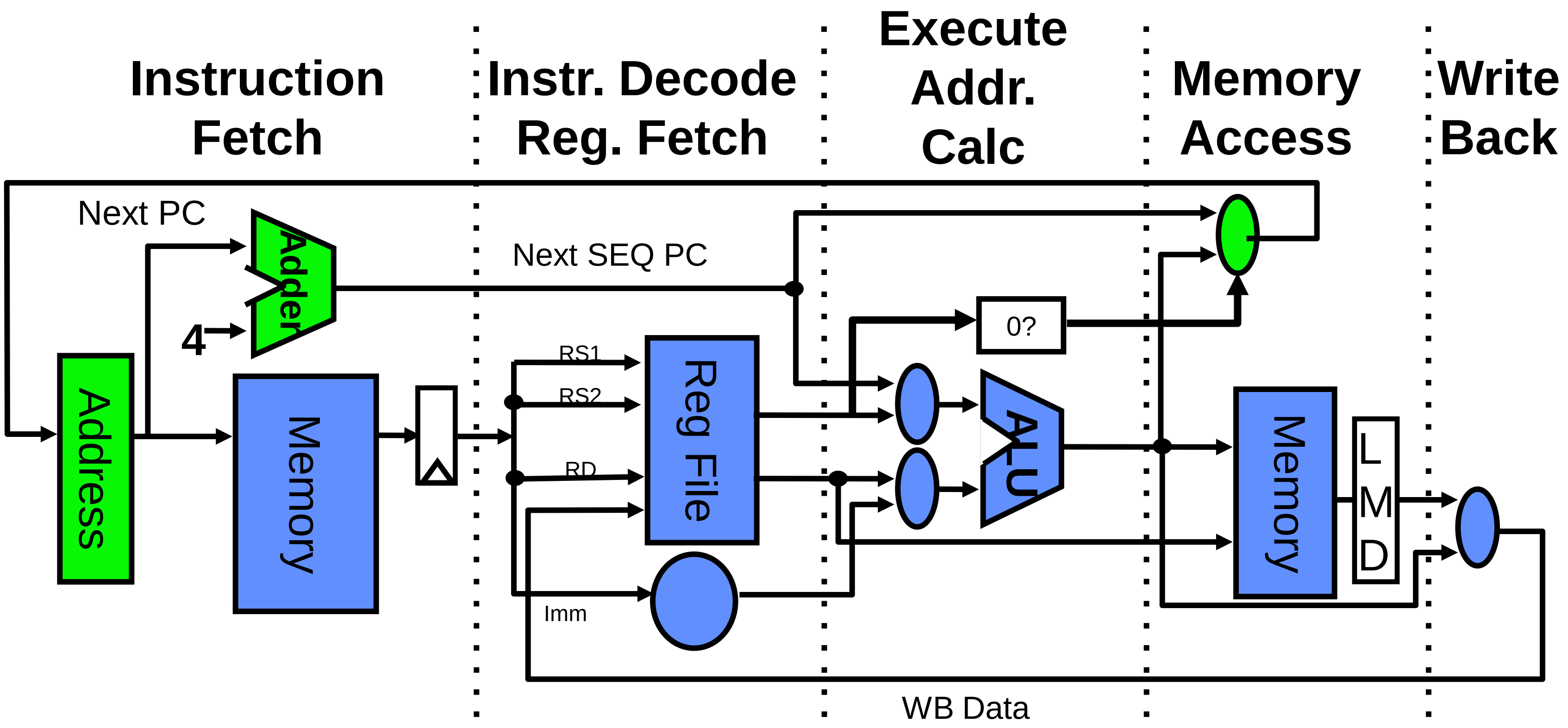


Pipelining Characteristics

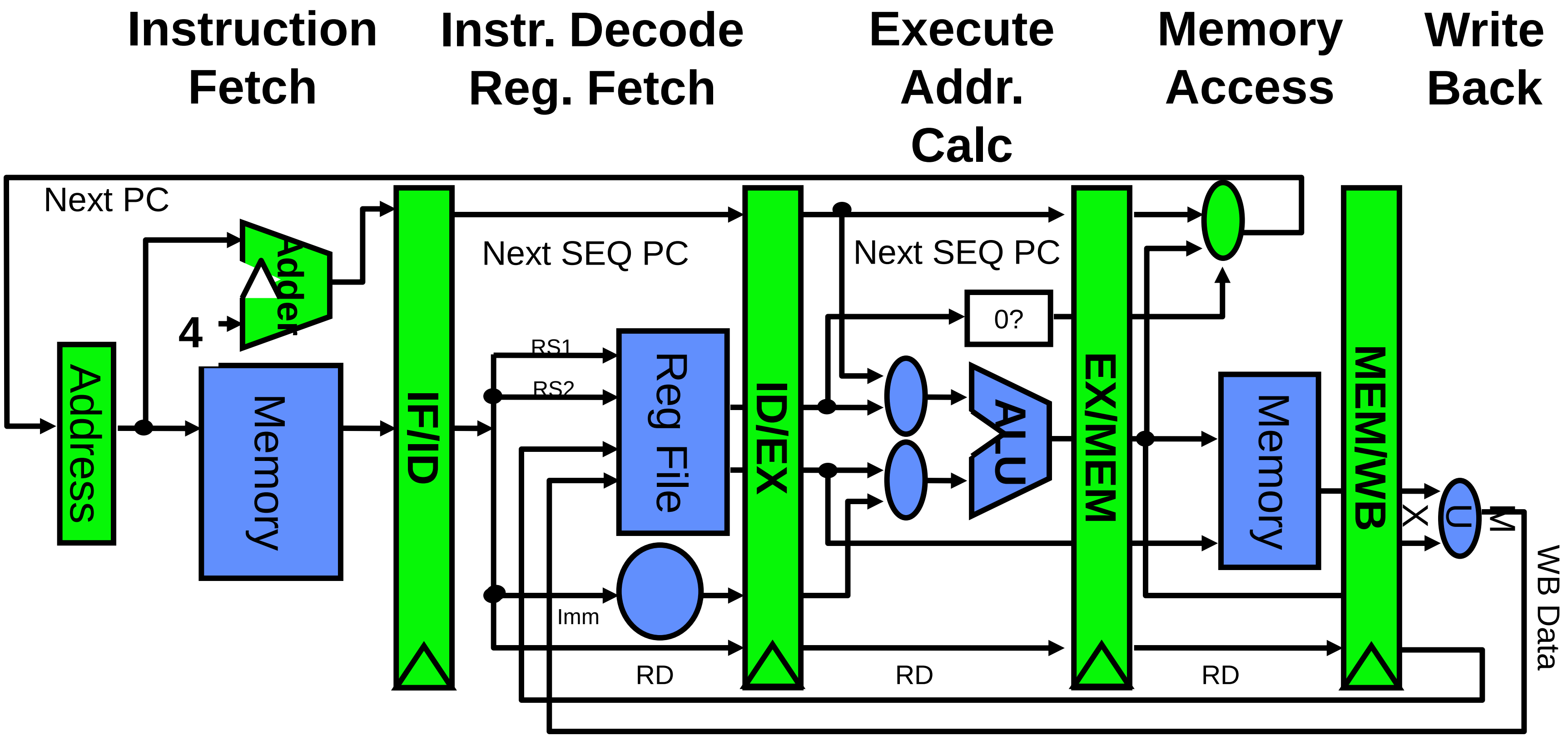


- Pipelining doesn't reduce latency of single task, it improves throughput of entire workload
- Pipeline rate limited by slowest pipeline stage
- Potential speedup = Number of pipe stages

Unpipelined RISC Data path

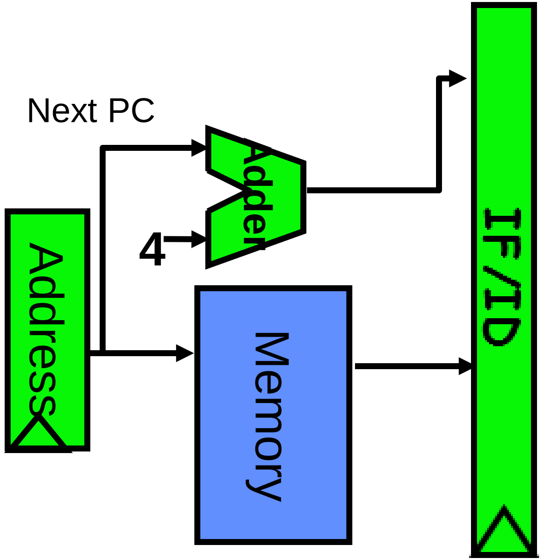


Pipelined RISC Data path



RISC MIPS Instruction Pipeline

- Each instruction can take at most 5 clock cycles
- **Instruction fetch cycle (IF)**
  - Based on PC, fetch the instruction from memory
  - Increment PC



RISC MIPS Instruction Pipeline

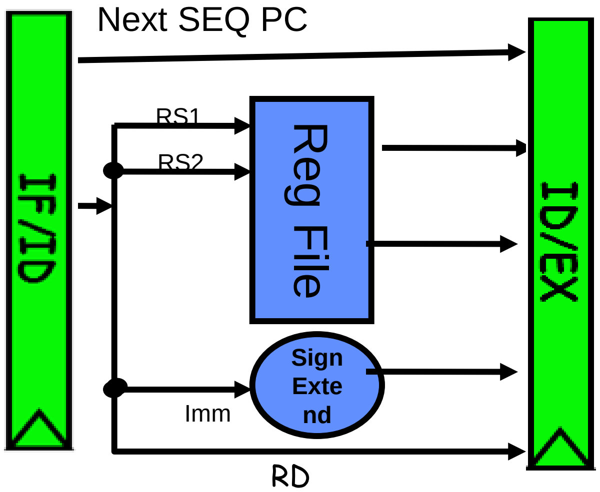
- **Instruction decode/register fetch cycle (ID)**
- Decode the instruction + register read operation
- Fixed field decoding

Ex: ADD R1,R2,R3 : A3.01.02.03

10100011 00000001 00000010 00000011

Ex: LW R1,8(R2) : AE.00.08.02

10101110 00000000 00001000 00000010



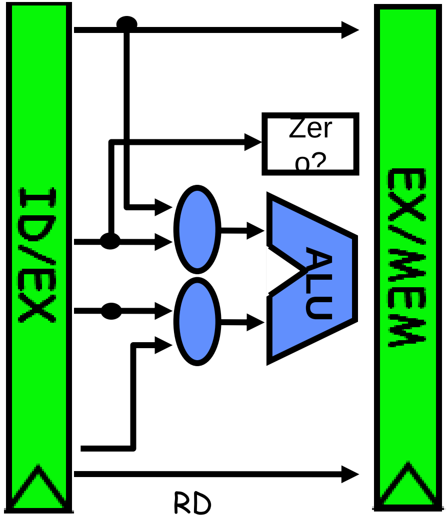
RISC MIPS Instruction Pipeline

- **Execution/Effective address cycle (EX)**
- Memory reference: Calculate the effective address

  LW R1, 8(R2) **EFF ADDR= [R2] +8**

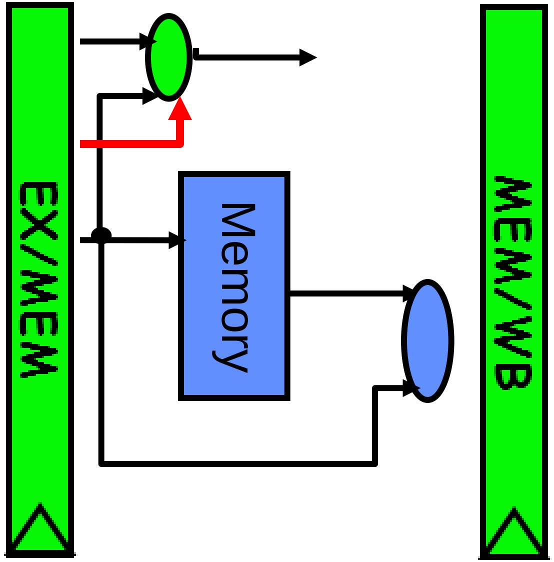
- Register-register ALU instruction

  ADD R1,R2,R2



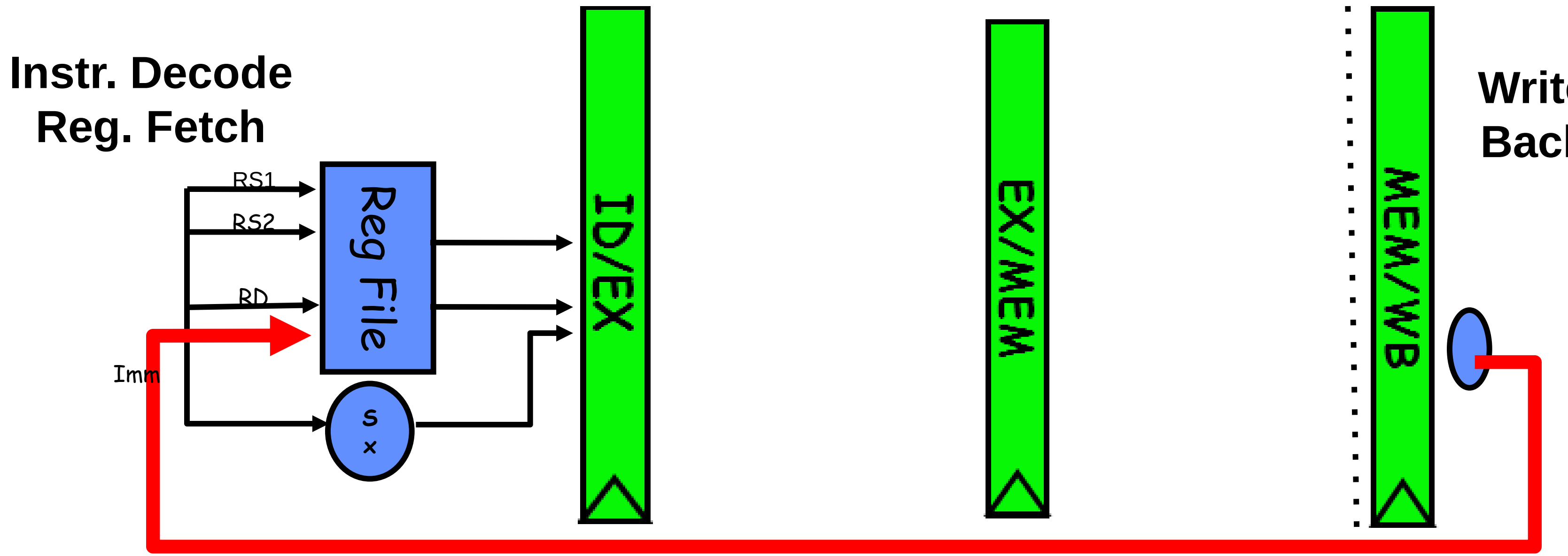
RISC MIPS Instruction Pipeline

- ❖ **Memory access cycle (MEM)**
  - ❖ Load from memory and store in register **[LW R1,8(R2)]**
  - ❖ Store the data from the register to memory**[SW R3,16(R4)]**

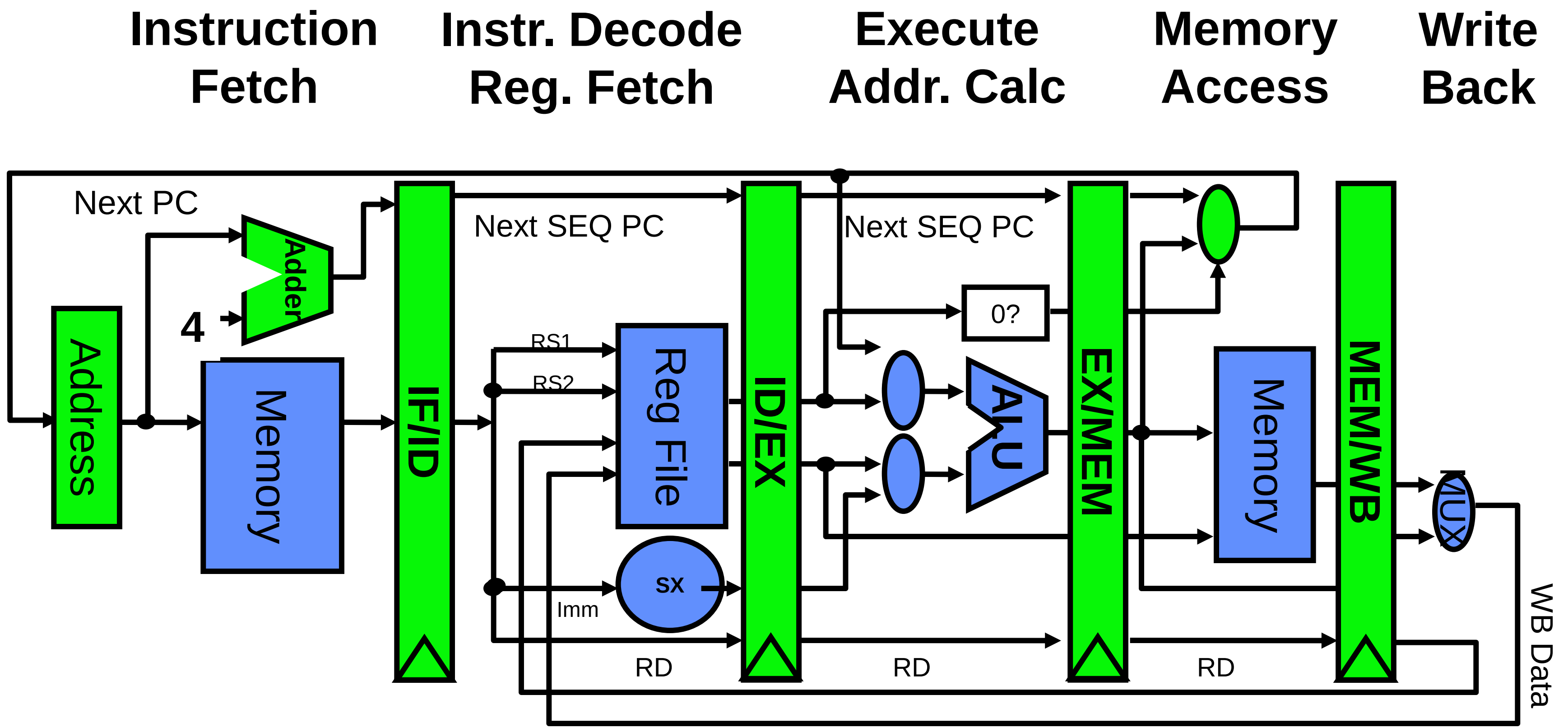


RISC Instruction Pipeline

- **Write-back cycle (WB)**
  - Register-register ALU instruction or load instruction
  - Write to register file **LW R1,8(R2) , ADD R1,R2,R3**



Pipelined RISC Data path



5 Steps of RISC Data path

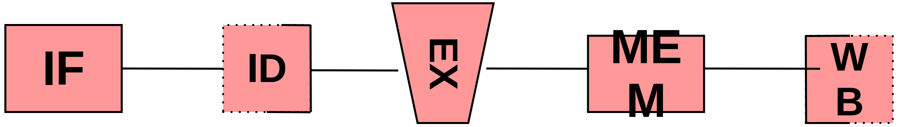
- Each instruction can take at most 5 clock cycles
- **Instruction fetch cycle (IF)**
  - Based on PC value, fetch the instruction from memory
  - Update PC
- **Instruction decode/register fetch cycle (ID)**
  - Decode the instruction + register read operation
  - Fixed field decoding
  - Equality check of registers
  - Computation of branch target address if any

5 Steps of MIPS Data path

- **Execution/Effective address cycle (EX)**
  - Memory reference: Calculate the effective address
  - Register-register ALU instruction
  - Register-immediate ALU instruction
- **Memory access cycle (MEM)**
  - Load instruction: Read from memory using effective address
  - Store instruction: Write the data in theregister to memory using effective address

5 Steps of MIPS Data path

- **Write-back cycle (WB)**
  - Register-register ALU instruction or load instruction
  - Write the result into the register file
- **Cycles required to implement different instructions**
  - Branch instructions – 4 cycles
  - Store instructions – 4 cycles
  - All other instructions – 5 cycles



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Pipelining Issues

- ❖ **Ideal Case: Uniform sub-computations**
  - ❖ The computation to be performed can be evenly partitioned into uniform-latency sub-computations
- ❖ **Reality: Internal fragmentation**
  - ❖ Not all pipeline stages may have the uniform latencies
- ❖ **Impact of ISA**
  - ❖ Memory access is a critical sub-computation
  - ❖ Memory addressing modes should be minimized
  - ❖ Fast cache memories should be employed

Pipelining Issues

- **Ideal Case : Identical computations**
  - The same computation is to be performed repeatedly on a large number of input data sets
- **Reality: External fragmentation**
  - Some pipeline stages may not be used
- **Impact of ISA**
  - Reduce the complexity and diversity of the instruction types
  - RISC architectures use uniform stage simple instructions

Pipelining Issues

- **Ideal Case : Independent computations**
  - All the instructions are mutually independent
- **Reality: Pipeline stalls – cannot proceed.**
  - A later computation may require the result of an earlier computation
- **Impact of ISA**
  - Reduce Memory addressing modes - dependency detection
  - Use register addressing mode - easy dependencies check

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